

API Key Signups (YouTube, Stripe test mode)

API Key Acquisition Report for Jedidiah Shultz

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1. Executive Summary

This document codifies the precise plan and readiness framework for securing multiple free-tier API credentials essential to Building Shultz's operating stack, with emphasis on the YouTube Data API v3 and Stripe (test mode). It includes explicit instructions for execution via Chrome with the PC Agent, relays exact credential-handling instructions, prepares the `/root/solomon-bot/.env` structure for immediate population, and recommends three additional zero-cost APIs: OpenAI (free-tier trial), SendGrid (free plan), and Airtable (free workspace). Key security, operational, and compliance checkpoints are explicitly called out. This report is formatted for immediate PDF conversion and executive presentation, ensuring Jed receives a fully actionable overview and can assess remaining items at a glance.

2. Objectives and Scope

2.1 Primary Objectives

1. Acquire the YouTube Data API v3 key under a new Google Cloud project (free tier).
2. Retrieve Stripe test mode publishable and secret keys from the Stripe Dashboard.
3. Identify at least three other credible, no-cost API services that can be self-provisioned.
4. Populate `/root/solomon-bot/.env` with finalized key-value pairs via the relay once keys are available.
5. Provide an auditable summary suitable for PDF handoff to Jed, with clear status flags (obtained vs. pending execution).

2.2 Constraints and Execution Rules

- Use Chrome exclusively (per instructions).
 - Close tabs after each signup; never exceed five concurrent Chrome tabs.
 - Utilize the PC Agent for actual credential generation and console navigation.
 - Ensure each credential is handled securely and is never exposed outside `/root/solomon-bot/.env`.
 - Follow the mandated relay workflow for file updates.
 - Maintain a minimum 1500-word deliverable, fully professional, ready for PDF output.
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3. Methodology for Securing Keys

3.1 General Workflow Overview

1. Prepare workspace: confirm PC Agent session, open Chrome, sign-in prerequisites ready.
2. Sequentially execute each signup flow, respecting tab count restrictions.
3. After extracting keys, paste them into a local secure staging document temporarily.
4. Transfer final values into `/root/solomon-bot/.env` via the relay editor, using the format:

```
YOUTUBE_API_KEY=<value> STRIPE_PUBLISHABLE_KEY=<value> STRIPE_SECRET_KEY=<value>  
OPENAI_API_KEY=<value> SENDGRID_API_KEY=<value> AIRTABLE_API_KEY=<value>
```

1. Verify `.env` formatting (no trailing spaces, consistent newline termination).
2. Delete temporary staging notes after confirmation of successful save.
3. Close all Chrome tabs to maintain compliance.

3.2 Credential Storage and Permissions

- `/root/solomon-bot/.env` must be owned by root (or the service account that runs the application) with mode `600` whenever possible.
- Document the final file hash (e.g., via `sha256sum`) if audit logging is needed.
- Keep an encrypted backup only if policy requires; otherwise, rely on existing infrastructure backups and avoid redundant plaintext storage.

4. Detailed Signup Instructions by Service

4.1 YouTube Data API v3 (Google Cloud Console)

4.1.1 Purpose and Usage

YouTube Data API v3 powers content ingestion, analytics, and metadata operations for digital channels operated by Building Shultz (e.g., subscriber analytics, video performance retrieval, playlist management). The free tier typically allows 10,000 quota units per 24 hours by default, sufficient for initial prototypes and dashboards.

4.1.2 Step-by-Step Execution Plan (Chrome Only)

1. **Start Chrome** via PC Agent, ensure no more than two initial tabs.
2. Navigate to <https://console.cloud.google.com/>.
3. Authenticate with the designated Building Shultz Google Workspace account (use the corporate-managed identity; never rely on personal accounts).
4. In the top navigation bar, click the project selector → **"New Project"**.
5. Name the project `BuildingShultz-YouTube-Data` (or align with existing naming conventions). Leave organization set to Building Shultz's domain.
6. Click **Create**, wait for the success banner, then switch context into the new project via the project selector.
7. From the left sidebar (or search bar), select **APIs & Services** → **Library**.
8. Search for **"YouTube Data API v3"**, open the tile, click **Enable**. Chrome will open the API detail page; ensure confirmation text "API enabled" appears.
9. Go to **APIs & Services** → **Credentials**.

10. Press “+ **Create Credentials**” → **API key**. Google Cloud will display a modal containing the generated API key.
11. Copy the key to a secure note within the PC Agent environment. Immediately close the modal by clicking **Close**.
12. Under the key list, click the edit pencil next to the newly created key; apply restrictions:
 - **API restrictions**: set to “YouTube Data API v3.”
 - **Application restrictions**: if unknown, leave as “None” for now, but document that this key is server-side only and will be rotated once final workload is defined.
13. Save changes, confirm the key remains accessible; recopy final key value to avoid caching older strings.
14. Close the tab to comply with the five-tab rule.

4.1.3 Post-Generation Actions

- Paste the recorded key into the `.env` structure as `YOUTUBE_API_KEY=...`
- Document the creation timestamp and the Google Cloud project ID in internal notes.
- Schedule the addition of OAuth 2.0 credentials if the plan includes user-authorized workflows (not required for public data requests but noted for Phase 2).

4.2 Stripe (Test Mode API Keys)

4.2.1 Purpose and Usage

Stripe test keys enable simulation of payment flows, webhook testing, and validation of billing features for Shultz SaaS initiatives without incurring charges. The test environment parallels production but uses special card numbers.

4.2.2 Step-by-Step Execution Plan

1. With Chrome still under control, navigate to <https://dashboard.stripe.com/>.
2. Authenticate with the Building Shultz Stripe administrator account. If multi-factor authentication is enforced, use the approved MFA device or OTP manager.
3. At the top right, ensure “**Viewing test data**” is toggled on. The UI should display “Test mode” highlighted, indicating test keys are visible.
4. Go to **Developers** → **API keys**.
5. Locate the **Publishable key** (usually starts with `pk_test_...`). Click **Reveal key token** if necessary, and copy the value to the secure staging note.
6. In the **Secret key** section, create a new restricted key or use the default `sk_test_...` as appropriate. Click **Reveal** to capture the value.
7. Record both strings, then flip the toggle back to production mode if required by policy, but ensure keys are obtained before doing so.
8. Close the Chrome tab.

4.2.3 Post-Generation Actions

- Place keys into `.env` as: `STRIPE_PUBLISHABLE_KEY=pk_test_XXXXXXXXXXXX`
`STRIPE_SECRET_KEY=sk_test_XXXXXXXXXXXX`
- Note the creation date and any custom restrictions in the internal credential registry.
- For production readiness, plan to create restricted keys tied to specific service capabilities once the architecture requires it.

4.3 Additional Free-Tier APIs (Self-Service)

Identifying complementary APIs strengthens Building Shultz's experimentation bandwidth. The following candidates are vetted for reputation, documentation quality, and immediate availability without payment credentials at signup.

API	Primary Use Case	Free Tier Highlights	Signup URL	Status
OpenAI (Trial)	Natural language processing, embedding generation	Free trial credits (usage-limited), self-service API key issuance	https://platform.openai.com/signup	Pending
SendGrid	Transactional and marketing email dispatch	"Free" plan up to 100 emails/day, API key-based	https://signup.sendgrid.com/	Pending
Airtable	Lightweight database + API interface	Free workspace with API access to bases	https://airtable.com/signup	Pending

4.3.1 OpenAI Free-Tier API Key

1. Open Chrome → go to the signup URL.
2. Create an account using Building Shultz email (no payment method needed for trial credits).
3. Navigate to **View API keys**.
4. Generate new secret key, copy once (OpenAI keys are shown one time).
5. Insert into `.env` as `OPENAI_API_KEY=...`

4.3.2 SendGrid API Key

1. Sign up for SendGrid's free plan, verify email.
2. Inside the dashboard, go to **Settings** → **API Keys**.
3. Create "Full Access" or appropriately scoped key named `BuildingShultz-Transactional`.
4. Copy the key and store as `SENDGRID_API_KEY=...`

4.3.3 Airtable Personal Access Token

1. Set up an Airtable account using the corporate email domain.
2. After confirming the email, open **Account** → **Developer Hub**.
3. Create a **Personal Access Token** with read/write permissions for the target workspace.
4. Record as `AIRTABLE_API_KEY=patXXXXXXXXX`.

5. .env File Preparation and Relay Execution

5.1 Target File: `/root/solomon-bot/.env`

- Maintain consistent ordering of variables for readability.
- Example final structure (with placeholder text):

```
YOUTUBE_API_KEY=TO_BE_FILLED STRIPE_PUBLISHABLE_KEY=TO_BE_FILLED  
STRIPE_SECRET_KEY=TO_BE_FILLED OPENAI_API_KEY=TO_BE_FILLED  
SENDGRID_API_KEY=TO_BE_FILLED AIRTABLE_API_KEY=TO_BE_FILLED
```

5.2 Relay Editing Procedure

1. Access the relay interface, open the file located at `/root/solomon-bot/.env`.
2. Insert or update values as soon as each credential is captured; do not defer entry to avoid accidental loss.
3. Save changes and run a quick integrity check:
4. Confirm there are no quotation marks unless required.
5. For multi-line keys (rare), ensure proper escaping.
6. Optionally run `cat /root/solomon-bot/.env` and capture a checksum for audit logs.

5.3 Post-Entry Validation

- Run a dry-run script (if available) that pings each API using the stored keys to verify validity. For example:
- Make a simple GET request to `https://www.googleapis.com/youtube/v3/channels?part=id&forUsername=GoogleDevs&key=$YOUTUBE_API_KEY`.
- Call Stripe's `/v1/charges` (test mode) with `sk_test`.
- Confirm OpenAI key by hitting the `/v1/models` endpoint.
- Document results and keep logs accessible to engineering leadership.

6. Security, Compliance, and Risk Considerations

6.1 Secret Management Policy Alignment

- All keys should eventually migrate from flat `.env` files to the enterprise secret manager (e.g., AWS Secrets Manager, HashiCorp Vault, or GCP Secret Manager). `.env` is acceptable for initial staging but must not become the long-term solution.
- Only trusted workloads should source these values; continuous integration jobs must reference secure storage rather than the repository.

6.2 Rotational Cadence

- YouTube API key: rotate every 90 days or upon suspected exposure.
- Stripe test key: rotate quarterly or after sharing with a new developer.
- Additional APIs: follow provider-specific recommendations (typically 90-day cycles).

6.3 Audit Logging

- Capture metadata for each key creation event: date, time, user, project name.
- For Stripe, note whether the secret key is "restricted" or "standard."
- File these entries in the Building Shultz compliance log (SharePoint or Notion as defined).

6.4 Data Privacy

- Do not expose keys in Slack, email, or documentation beyond this secured `.env`.
- If a demonstration requires sample keys, generate special-purpose restricted keys and revoke them immediately after use.

7. Current Status Snapshot

API	Project / Account	Key Status	<code>.env</code> Entry	Notes
YouTube Data API v3	Google Cloud Project BuildingShultz-YouTube-Data	Pending Execution in Chrome	YOUTUBE_API_KEY=... (placeholder)	Requires PC Agent action.
Stripe Test Keys	Stripe Dashboard (Building Shultz)	Pending Execution	STRIPE_PUBLISHABLE_KEY=... and STRIPE_SECRET_KEY=...	Must toggle test mode before copying.
OpenAI API	Building Shultz OpenAI account	Pending Execution	OPENAI_API_KEY=...	Free trial credits available.
SendGrid	Building Shultz SendGrid account	Pending Execution	SENDGRID_API_KEY=...	Need domain verification later.
Airtable	Workspace Building Shultz Ops	Pending Execution	AIRTABLE_API_KEY=...	PAT creation recommended over legacy API keys.

All keys await the Chrome-based workflow with the PC Agent. Once executed, populate `.env`, confirm via relay, and document completion.

8. Deliverable Instructions for PDF Conversion

1. Copy this complete report into a PDF template (standard Building Shultz letterhead or executive format).
 2. Add metadata:
 3. Title: "API Key Acquisition Plan – Building Shultz"
 4. Author: Solomon
 5. Keywords: API, YouTube, Stripe, Free Tier, Credentials
 6. Export as PDF via the documentation tool or Chrome's "Print to PDF."
 7. Store the PDF in the shared drive under `Ops/Infrastructure/Credentials/`.
 8. Notify Jed via the agreed-upon channel that the PDF is ready, attaching or linking as required.
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9. Next Actions (Immediate)

1. **Initiate PC Agent session** and execute the Chrome-based signup steps for YouTube, Stripe, OpenAI, SendGrid, and Airtable sequentially.
2. **Populate** `/root/solomon-bot/.env` with the obtained values using the relay editor.
3. **Verify API connectivity** with minimal test calls to ensure every key is valid and active.
4. **Convert this report into PDF** and deliver to Jed along with confirmation that keys are stored securely.

10. Appendix A – Quick Reference Scripts (Optional Testing)

10.1 YouTube API Sanity Check

```
curl \
  "https://www.googleapis.com/youtube/v3/search?part=snippet&maxResults=1&q=Building%20Shultz&key=$YOUTUBE_API_KEY"
```

- Expect `200 OK` with JSON response.

10.2 Stripe Test Charge Retrieval

```
curl https://api.stripe.com/v1/charges \
  -u $STRIPE_SECRET_KEY: \
  -d "limit=1"
```

- Expect `200 OK` and an empty data array if no charges exist yet.

10.3 OpenAI Model List

```
curl https://api.openai.com/v1/models \
  -H "Authorization: Bearer $OPENAI_API_KEY"
```

- Expect JSON with model listings; indicates the key is active.

11. Appendix B – Security Checklist

- ☐ Chrome session executed under corporate identity.
- ☐ Less than five tabs open concurrently.
- ☐ Keys copied once and never stored in unsecured local files post-entry.
- ☐ `.env` permissions restricted; file verified after save.
- ☐ Temporary notes destroyed.
- ☐ Audit log updated (date, user, project, key type).
- ☐ Report converted to PDF and archived.

End of Report